

Comparison of Random Forest and Support Vector Machine algorithms for the prediction of clinical endpoints in neuroblastoma.

Anneke Prins, MSc Bioinformatics, University of Birmingham.

Abstract

High throughput methods like microarrays provide a rich resource for the identification of predictive biomarkers that can aid in the prognosis of heterogenous diseases like neuroblastoma. Machine learning algorithms such as random forest and support vector machine provide two strategies for the identification of a small subset of predictive biomarkers. Here I have shown the ability of these models to identify several features that are predictive for 4 clinical endpoints, and compared their accuracy. I conclude that random forest is the more practical model to use, and that several biomarkers are predictive of more than one clinical endpoint, which could be extremely useful in a clinical setting.

1. Introduction

With the advent of high-throughput screening, the usage of gene expression profiles have been used in predictive modelling of various diseases (Pittman et al, 2004; Babu and Suresh, 2013; Van Dam et al, 2018; Lee and Lee, 2020; Qian et al, 2021). While most current studies use RNA-Seq platforms, the availability of existent microarray data represent a rich resource for the identification of biomarkers (Kim et al, 2024). The goal is to identify the smallest set of genes that can achieve good predictive performance. Because microarray data is high dimensional, machine learning approaches such as random forest (RF) and support vector machine (SVM) allow the identification of smaller subsets of meaningful features for the prediction of outcomes. The RF algorithm is one of the most popular classifiers in use (Fernández-Delgado et al, 2014; Acharjee et al, 2020) It is an ensemble method that uses a collection of classification trees, where each tree uses a bootstrap of the sample data and a random subset of the variables at each decision node (Díaz-Uriarte and Andrés, 2006). An alternative approach is found in the SVM algorithm which aims to find a boundary that maximally separates different classes (Bansal et al, 2022). While it has been widely used in image, hypertext and text segregation, it has also been applied in predictive modelling in different diseases (Ali et al, 2020; Biswas et al, 2022; Nugroho et al, 2022). SVM performs as well as RF in classification (Wainberg et al, 2016), and therefore deserves scrutiny as a method in disease prediction.

Neuroblastoma is a rare disease that mostly affects children under the age of five (Nong et al, 2025). The disease has a very heterogenous clinical course, making it difficult to predict the outcome of disease at the point of diagnosis. While age at diagnosis and MYCN gene amplification provide information for prognosis, these have limited application since MYCN amplification represent only 22% of all neuroblastoma tumours (Newman and Nuchtern, 2016). Therefore the aim of this study was to apply two modelling algorithms (RF and SVM) to identify biomarkers for the prediction of clinical endpoints in neuroblastoma, using microarray and clinical data (Zhang et al, 2015). The objectives were to identify a small set of predictive biomarkers and to compare the performance of two classifiers in this disease. The hypothesis was that some biomarkers would be predictive of a clinical endpoint using either of the algorithms, or both.

2. Methodology

An overview of the approach used in this study is provided in Figure 1.

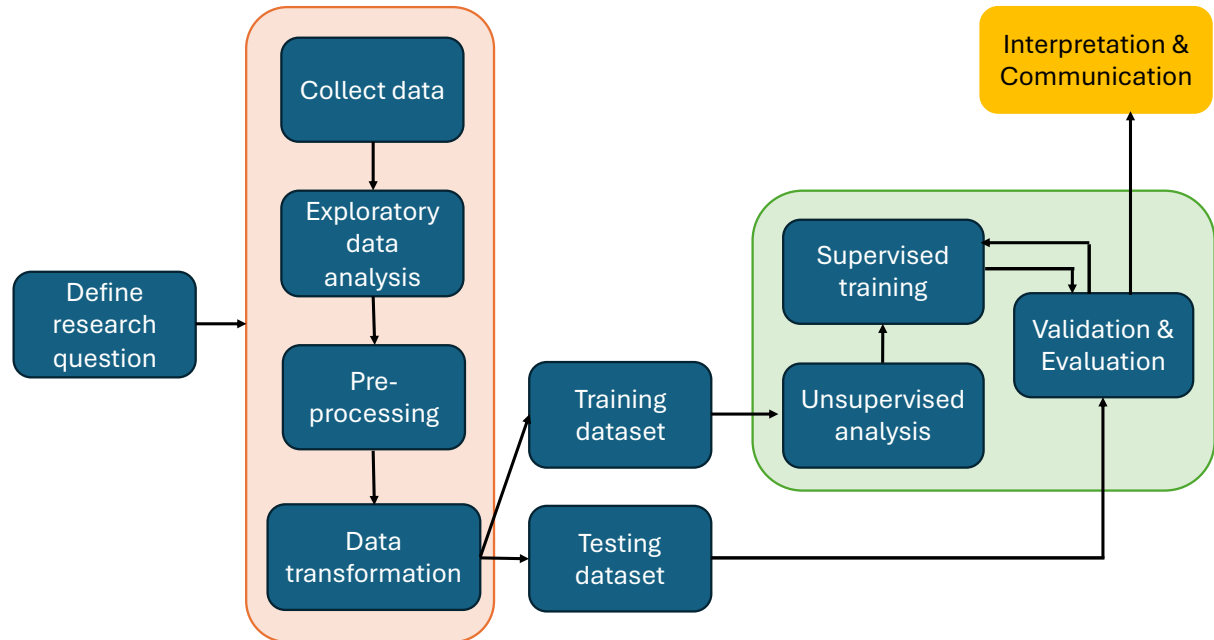


Figure 1. Strategy to train a predictive model for several clinical endpoints in neuroblastoma.

2.1 Data sets

Gene expression data was obtained from Zhang et al (2015) who used custom 4x44k oligonucleotide arrays (Agilent Technologies) to generate gene expression profiles from 498 primary neuroblastomas. Clinical data was obtained from the same source (Table 1).

Table 1. Clinical data. Clinical endpoints for prediction are highlighted in blue. Other clinical features were incorporated in the predictive model.

Feature:	Gender	Age	clinico.genetic.subgroup	MYCN.status	INSS.Stage	HighRisk	Progression	DeathFromDisease
<i>Definition</i>	Gender of patient	Age at diagnosis in days	Patient subgroup with same genetic and clinical characteristics	Amplification status of the MYCN proto-oncogene	Disease stage	Clinically high risk neuroblastoma	Occurrence of tumour progression	Occurrence of death from disease
<i>Value type</i>	Categorical	Integer	Categorical	Categorical	Mixed	Categorical	Integer	Integer
<i>Values</i>	M, F	0 to 8983	MNA, ST1, ST4, ST4S	1, Amp	1, 2, 2a, 2b, 3, 4, 4S	LR, HR	0, 1	0,1

2.2 Exploratory Data Analysis

Distribution of different categories in the clinical data was analysed using frequency tables and graphs. Summary statistics were obtained for gene data, and the distribution of the first 10 probes in the data set investigated using histograms. Outliers were identified by performing principle component analysis (PCA) on the gene data as implemented in SciKit Learn (sklearn, version 1.6.0) in Python (version 3.9.21).

2.3 Preprocessing

Probes with no expression data were removed from the set and probe names standardised by the removal of extraneous tags. Microarray data was subset to reflect samples present in the clinical data. Categorical predictors were re-coded as binary (0/1) for two outcomes or one-hot encoded for multi-category outcomes using the OneHotEncoder preprocessing tool from sklearn.

The data were separated into input features (microarray probe data, gender, age, clinico-genetic subgroup and MYCN status) and output features (INSS stage, risk, progression and death from disease), and then split into train and test sets before imputation or scaling. The data was split 80/20 train/test using stratification on the 'death from disease' outcome variable, using the train_test_split tool (sklearn). Missing values were imputed as the median value of the feature using SimpleImputer and data was scaled using StandardScaler (sklearn).

2.4 Unsupervised Analysis

Unsupervised analysis of train data was done by PCA as implemented in SciKit-Learn.

2.5 Model training

Model training and evaluation was done in a nested approach, with the aim of predicting four clinical endpoints (INSS Stage, risk, progression and death from disease). In the first instance, the train data set was trained to predict death from disease using a stratified k-fold cross validation approach, with 5 folds and shuffling between fold selection (StratifiedKFold in sklearn). Two models were applied, namely SVM (SVC in sklearn) and RF (RandomForestClassifier sklearn). The number of features included in the classification was varied between 30 and 200, and feature selection was performed using both k-best selection (SelectKBest) and selecting the top 1 percentile features (SelectPercentile) based on the analysis of variance F-value (ANOVA F-test). Models were trained using these features for each fold, and the features that were common between both approaches were used to subset the train and test data sets before hyperparameter tuning. Hyperparameter tuning was performed using grid search cross validation (GridSearchCV in sklearn, cross-validation = 3). The trained models were evaluated on the untouched test set to return accuracy and F1 scores, and to plot ROC-AUC curves and Precision-Recall curves using matplotlib (version 3.3.4).

3. Results

There were 373 clinical samples with a mixture of numerical and categorical content (Table. 1 and Supplementary tables S1 and S2). There were no missing values in this data set. There was a fairly even split between males and females (218 to 155 respectively), but an imbalance in the number that died from the disease compared to those that didn't (277 compared to 96). There was also a greater proportion of males that did not die from disease compared to females (Fig. 2a). There was also an imbalance between high risk and low risk when comparing genders (Fig. 2b). Most samples were either in INSS stage 1 or 4 (Fig.2), and a minority of samples showed amplification in the MYC gene (Fig. 2d)

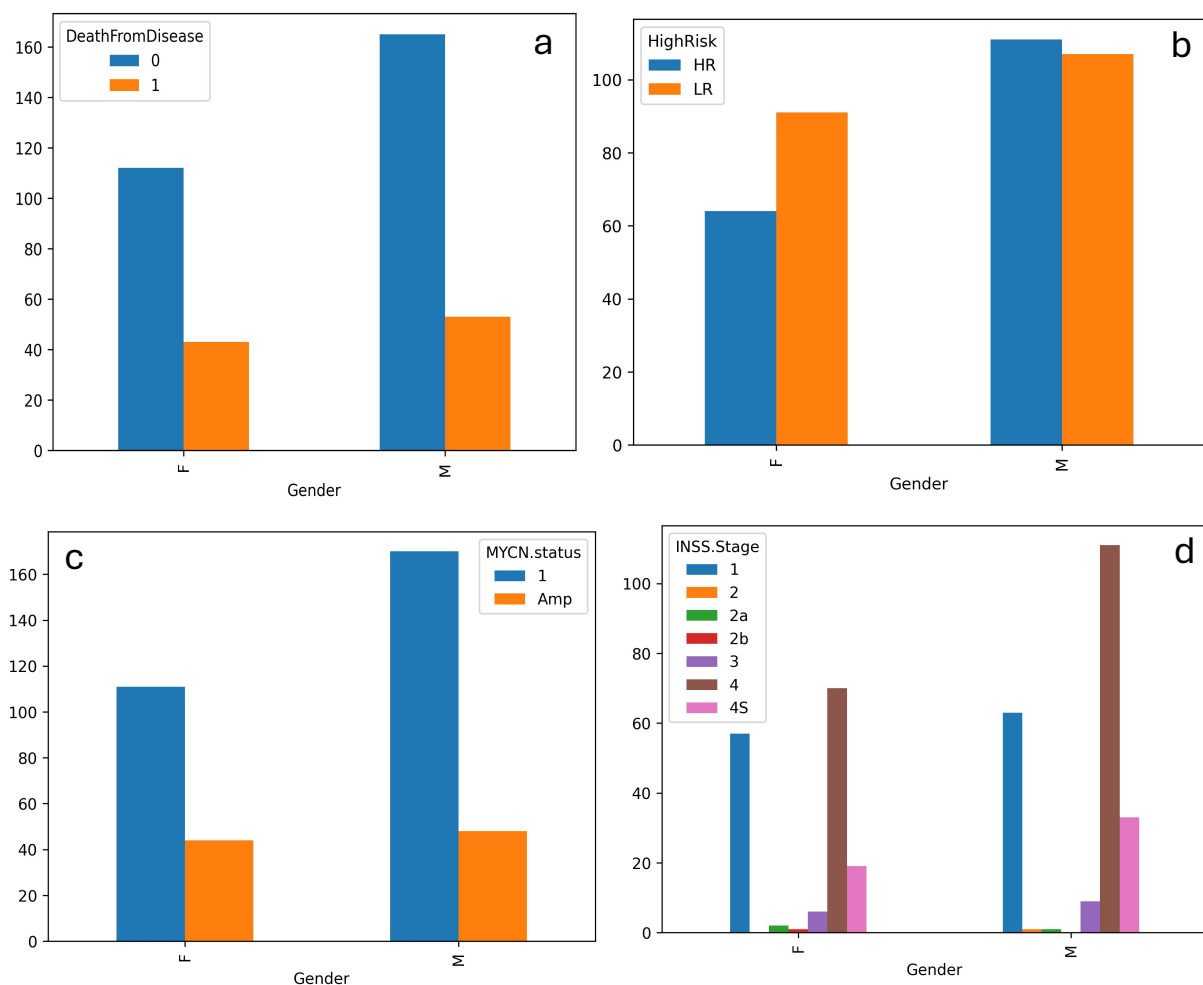


Figure 2. Frequencies of different classes in clinical data.

In the microarray data, there was a substantial number of missing values in each column, with most probes missing at least 5700 values. In addition, of the 498 probe values, 373 corresponding to the clinical features were retained for further analysis (Figure 3).

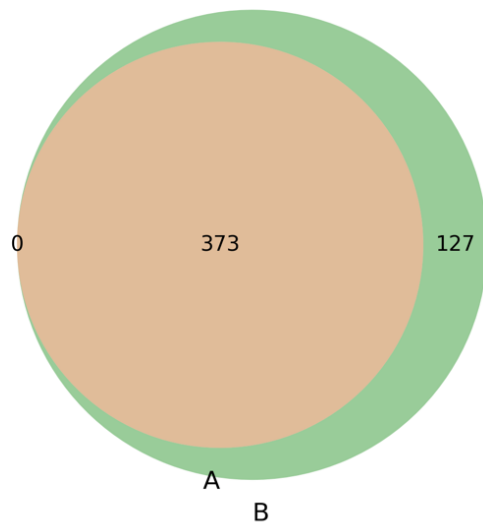


Figure 3. The number of samples shared between clinical features (A) and microarray expression data (B).

Of the 44,708 probes in the microarray data, only 7005 were related to known genes, and 1400 of these genes were related to more than one probe. Removing probes with no content at all decreased the dataset to 40,237 probes.

Distribution analysis of the first 10 probes in the gene set revealed some skewing in the data (Fig. 4). While RF isn't sensitive to skewing, SVM could be and therefore standardising was applied to the data set.

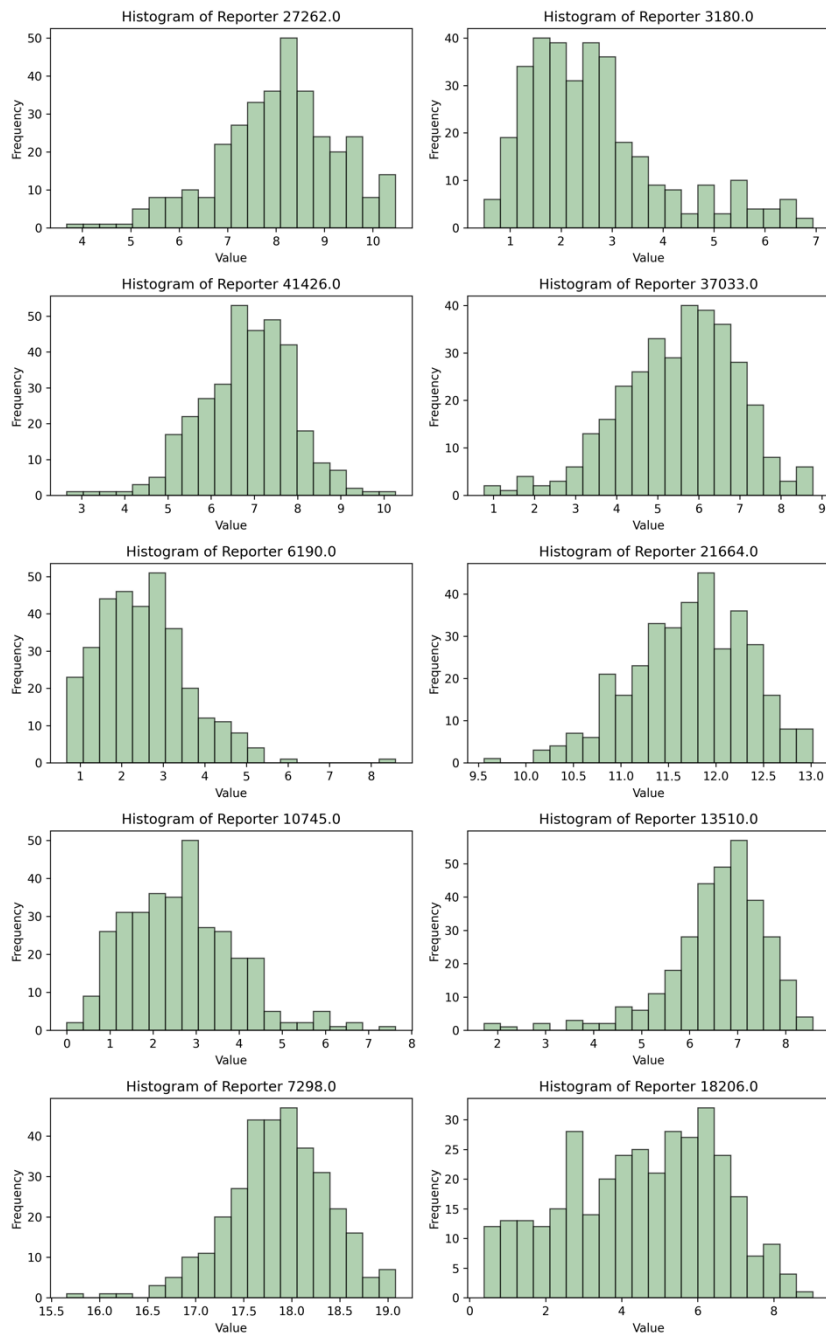


Figure 4. Distribution of gene expression data for 10 probes in the dataset.

An attempt was made to identify outliers in the gene data using PCA. The result revealed a fairly large, open cluster with no apparent patterns or outliers in it (Figure S1).

After splitting the train and test data, imputation and standardisation was done followed by unsupervised analysis using PCA (Fig. 5). This approach was not able to separate patients who died from disease, but showed more promise when analysing for risk status ('HighRisk'). However, PCA1 explained over 97% of the overall variation and was dominated by the contribution from the Age feature (Supplementary Table S3).

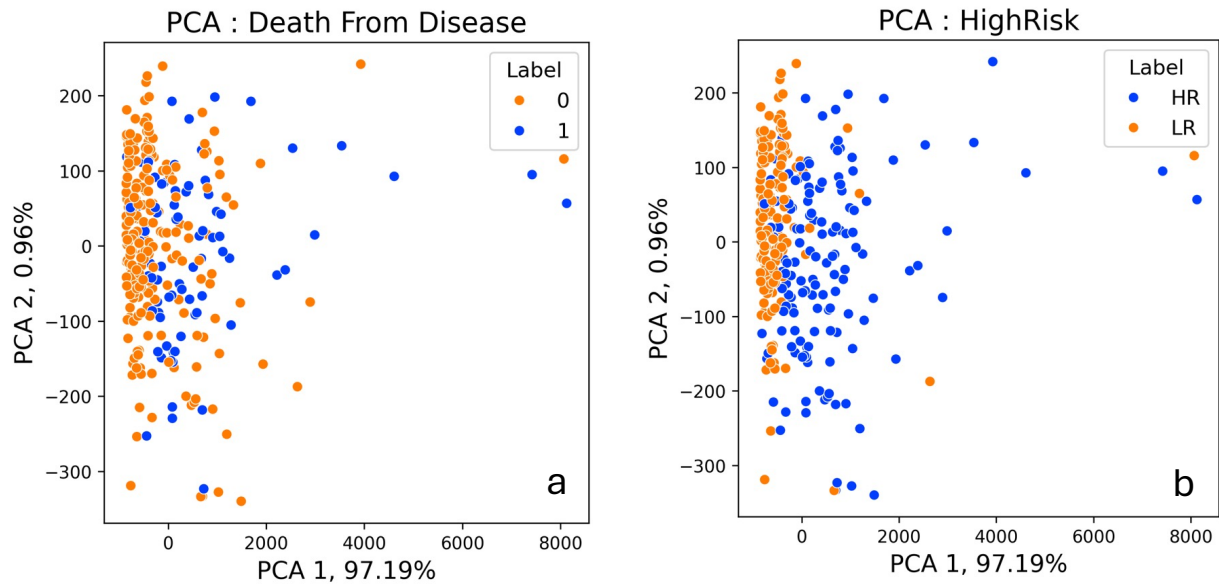


Figure 5. PCA analysis of microarray data. Samples coloured according to death from disease (a, 0 = no death; 1 = death) and risk status (b).

Model training was performed using RF and SVM. After feature selection and hyperparameter tuning RF had a slightly higher accuracy than SVM to predict death from disease when applied to the untouched test set (Table 2).

Table 2. Model parameters tested and selected for final trained model, as well as final accuracy score for RF and SVM algorithms. Clinical endpoint: death from disease.

<i>Model</i>	No. of features tested	No. of features used	Hyperparamter	Values tested	Value used	Final accuracy
<i>RF</i>	30	10	n_estimators	50, 100, 200	50	0.77
			max_depth	None, 10, 20	20	
			min_samples_split	2, 5, 10	10	
<i>SVM</i>	30	10	C	0.1, 1, 10	0.1	0.73
			kernel	rbf, poly, sigmoid	sigmoid	

The features selected for the ‘death from disease’ predictive model were all gene probes (Fig. 6) with probe 28655 having the highest importance.

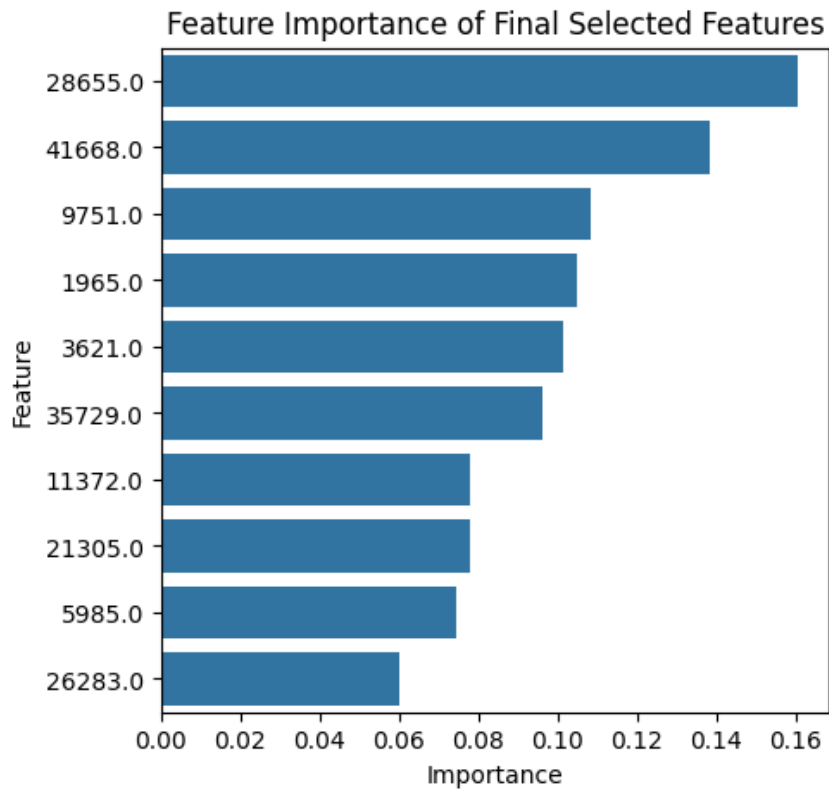


Figure 6. Features selected as most important for training the model to predict death from disease.

Evaluation was also performed using ROC and precision-recall (PR) curves (Fig. 7). While accuracy and ROC AUC returned high scores, F1 and PR AUC did not (Table 3).

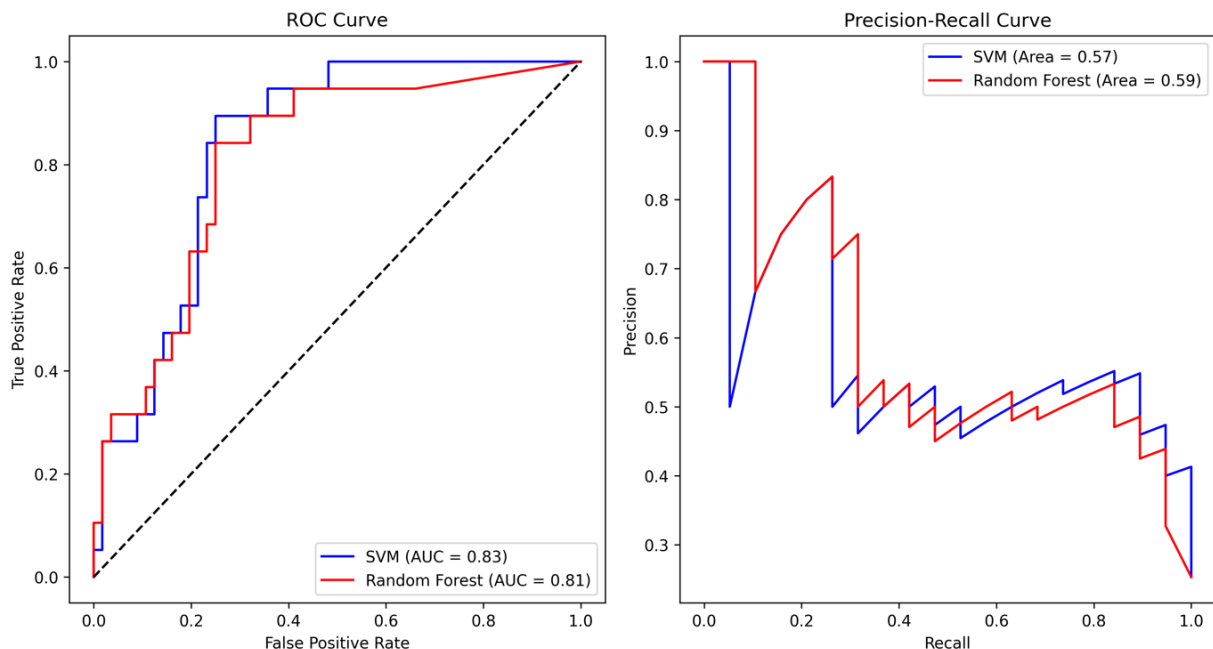


Figure 7. Performance evaluation using ROC-AUC and PR-AUC curves for predicting death from disease.

Table 3. Summary of performance scores obtained after training two models to predict death from disease.

Model	RF	SVM
Accuracy	0.77	0.73
F1 Score	0.51	0.50
ROC AUC	0.81	0.83
PR AUC	0.59	0.57

Of the 10 features used in the trained model, only one (35729) was associated with a gene, namely CTF1.

Other clinical endpoints were also used as output variables. For these models, accuracy score and selected features were collected (Table 4). A single predictive probe was shared between ‘death from disease’ and ‘progression’, while 5 probes were shared between ‘risk’ and ‘INSS stage’ prediction (Table 5). The model predicting INSS stage was the only one that incorporated categorical predictors, namely ST1, ST4, and ST4S. Final model configurations for clinical endpoints that aren’t ‘death from disease’ are available in the Appendix (Data S1).

Table 4. Accuracy scores of random forest (RF) and support vector machine (SVM) predictors for 3 clinical endpoints.

Endpoint	Accuracy score	
	RF	SVM
High Risk	0.92	0.88
Progression	0.71	0.73
INSS Stage	0.93	0.93

Table 5. Common features selected for prediction of 4 clinical endpoints using RF and SVM.

	Clinical endpoint				
	Death from disease	High Risk	Progression	INSS stage	Gene ID
Non-probe predictor	-	-	-	ST1, ST4S, ST4	-
Predictive probes (shared)	-	2731	-	2731	-
	-	9458	-	9458	-
	-	14867	-	14867	-
	-	23224	-	23224	-
	-	34728	-	34728	ADAMTS5
	28655	-	28655	-	-
Unique predictive probes with gene IDs	35729	-	-	-	CTF1
	-	34105	-	-	MTMR9LP
	--	45207		-	SLC5A4-AS1 CPSF1
	-	-	-	39434	BCDIN3D
	-	-	-	35733	GPNMB
Unique probes (no gene IDs)	1965	1995	4881	2396	-
	3621	3150	17520	8310	-
	5985	23272	-	10772	-
	9751	32099	-	22495	-
	11372	-	-	44212	-
	41668	-	-	-	-
	21305	-	-	-	-
	26283	-	-	-	-

4. Discussion

Out of the 44,000+ probes available on the microarray, 40,237 were used for training. Missing values were imputed using the median of the probe values to avoid affecting the variance and central tendency of the skewed data. Imputation was performed after the train/test split to avoid data leakage (i.e. the test set was treated as if it was sourced independently, with its own distribution and parameters).

Unsupervised analysis showed that the clinical endpoints could not be clustered by PCA and that PC1 explained over 97% of variance. The only endpoint that showed some separation was ‘HighRisk’, where age at diagnosis described 0.99% of the variation, with the rest of the probes having smaller contributions. It is already known that age at diagnosis holds prognostic value (Newman and Nuchtern, 2016), and so this approach would not be useful for more informed prediction.

When considering the clinical endpoint of ‘death from disease’, RF outperformed SVM slightly. Since RF is less computationally expensive than SVM, this would be the preferred algorithm in future. However, neither model did a good job of predicting the minority class (death from disease = 1) as can be seen from the F1 score and PR-AUC. While accuracy can be high because the model predicts the majority class correctly most of the time, the F1 score takes into consideration both precision and recall, and thereby penalises models that favour the majority class while ignoring the minority class. Therefore it would be prudent to re-test model performance with a data set that uses Synthetic Minority Over-sampling Technique (SMOTE) and/or Tomek Link (Swana et al, 2022).

RF and SVM were also compared for the prediction of ‘risk’, ‘progression’ and ‘INSS Stage’. Accuracy varied, with RF scoring high in HighRisk (0.92), lower than SVM in Progression (0.71) and high but similarly to SVM in INSS Stage (0.93). This illustrates that the appropriate model could depend on the endpoint being predicted and that the testing of more than one algorithm could return better results.

While many features were uniquely selected for a given clinical endpoint, ‘High Risk’ and ‘INSS stage’ shared 5 predictive probes, one of which (ADAMTS5) is a metalloproteinase that has been linked to glioblastoma (Nakada et al, 2005). Probe 35729, which relates to the gene CTF1, was uniquely linked to ‘death from disease’. The CTF1 gene encodes a cytokine from the interleukin-6 family. It causes cardiac myocyte hypertrophy and activates leukemia inhibitory factor/gp130 receptor complex (Pennica et al, 1996), which could have a functional role in cell proliferation during tumorigenesis. Other probes predictive for ‘risk’ and ‘INSS Stage’ were associated with a pseudogene (MTMR9LP; Mammalian Gene Collection Team, 2002), a non-coding RNA related to liver cancer (SLC5A4-AS1; Zeng et al, 2023), a methyltransferase predictive of prognosis in breast cancer (BCDIN3D; Yao et al, 2016) and a transmembrane protein that is enriched on the cell surface of cancer cells (GPNMB; Taya and Hammes, 2018).

5. Conclusion

In conclusion, this study has identified several features that can be used for the prediction of 4 clinical endpoints in neuroblastoma. However, this has to be taken with caution as the models applied were affected by imbalances in the available data set. In addition, there is evidence for RF being as good as SVM in model training. The current study is limited by the fact that several probes had to be removed during pre-processing and useful features could have been lost in the process. Future research could include over- or under sampling of the data to improve the strength of the model. The biomarkers could be validated in an independent study, and those linked to genes could be tested for a causative link which might lead to targeted therapies.

6. References

Acharjee, A., Larkman, J., Xu, Y., Cardoso, V.R. and Gkoutos, G.V., 2020. A random forest based biomarker discovery and power analysis framework for diagnostics research. *BMC medical genomics*, 13, pp.1-14.

- Ali, L., Wajahat, I., Amiri Golilarz, N., Keshtkar, F. and Bukhari, S.A.C., 2021. LDA–GA–SVM: improved hepatocellular carcinoma prediction through dimensionality reduction and genetically optimized support vector machine. *Neural Computing and Applications*, 33, pp.2783-2792.
- Babu, G.S. and Suresh, S., 2013. Parkinson's disease prediction using gene expression–A projection based learning meta-cognitive neural classifier approach. *Expert Systems with Applications*, 40(5), pp.1519-1529.
- Bansal, M., Goyal, A. and Choudhary, A., 2022. A comparative analysis of K-nearest neighbor, genetic, support vector machine, decision tree, and long short term memory algorithms in machine learning. *Decision Analytics Journal*, 3, p.100071.
- Biswas, N., Uddin, K.M.M., Rikta, S.T. and Dey, S.K., 2022. A comparative analysis of machine learning classifiers for stroke prediction: A predictive analytics approach. *Healthcare Analytics*, 2, p.100116.
- Díaz-Uriarte, R. and Alvarez de Andrés, S., 2006. Gene selection and classification of microarray data using random forest. *BMC bioinformatics*, 7, pp.1-13.
- Fernández-Delgado, M., Cernadas, E., Barro, S. and Amorim, D., 2014. Do we need hundreds of classifiers to solve real world classification problems?. *The journal of machine learning research*, 15(1), pp.3133-3181.
- Kim, W.J., Choi, B.R., Noh, J.J., Lee, Y.Y., Kim, T.J., Lee, J.W., Kim, B.G. and Choi, C.H., 2024. Comparison of RNA-Seq and microarray in the prediction of protein expression and survival prediction. *Frontiers in Genetics*, 15, p.1342021.
- Lee, T. and Lee, H., 2020. Prediction of Alzheimer's disease using blood gene expression data. *Scientific reports*, 10(1), p.3485.
- Mammalian Gene Collection (MGC) Program Team*, 2002. Generation and initial analysis of more than 15,000 full-length human and mouse cDNA sequences. *Proceedings of the National Academy of Sciences*, 99(26), pp.16899-16903.
- Miecznikowski, J.C., Sill, M. and Miller, A., 2024. An introduction to biomarkers in translational research (2023). In *Modern Inference Based on Health-Related Markers* (pp. 163-186). Academic Press.
- Nakada, M., Miyamori, H., Kita, D., Takahashi, T., Yamashita, J., Sato, H., Miura, R., Yamaguchi, Y. and Okada, Y., 2005. Human glioblastomas overexpress ADAMTS-5 that degrades brevican. *Acta neuropathologica*, 110, pp.239-246.
- Newman, E.A. and Nuchtern, J.G., 2016, October. Recent biologic and genetic advances in neuroblastoma: Implications for diagnostic, risk stratification, and treatment strategies. In *Seminars in pediatric surgery* (Vol. 25, No. 5, pp. 257-264). WB Saunders.
- Nong, J., Su, C., Li, C., Wang, C., Li, W., Li, Y., Chen, P., Li, Y., Li, Z., She, X. and Yuan, Z., 2025. Global, regional, and national epidemiology of childhood neuroblastoma (1990–2021): a statistical analysis of incidence, mortality, and DALYs. *EClinicalMedicine*, 79.

- Nugroho, K.S., Sukmadewa, A.Y., Vidiyanto, A. and Mahmudy, W.F., 2022. Effective predictive modelling for coronary artery diseases using support vector machine. *IAES International Journal of Artificial Intelligence*, 11(1), p.345.
- Pennica, D., Swanson, T.A., Shaw, K.J., Kuang, W.J., Gray, C.L., Beatty, B.G. and Wood, W.I., 1996. Human cardiotrophin-1: protein and gene structure, biological and binding activities, and chromosomal localization. *Cytokine*, 8(3), pp.183-189.
- Pittman, J., Huang, E., Dressman, H., Horng, C.F., Cheng, S.H., Tsou, M.H., Chen, C.M., Bild, A., Iversen, E.S., Huang, A.T. and Nevins, J.R., 2004. Integrated modeling of clinical and gene expression information for personalized prediction of disease outcomes. *Proceedings of the National Academy of Sciences*, 101(22), pp.8431-8436.
- Qian, Y., Daza, J., Itzel, T., Betge, J., Zhan, T., Marmé, F. and Teufel, A., 2021. Prognostic cancer gene expression signatures: current status and challenges. *Cells*, 10(3), p.648.
- Swana, E.F., Doorsamy, W. and Bokoro, P., 2022. Tomek link and SMOTE approaches for machine fault classification with an imbalanced dataset. *Sensors*, 22(9), p.3246.
- Taya, M. and Hammes, S.R., 2018. Glycoprotein non-metastatic melanoma protein B (GPNMB) and cancer: a novel potential therapeutic target. *Steroids*, 133, pp.102-107.
- Van Dam, S., Vosa, U., van der Graaf, A., Franke, L. and de Magalhaes, J.P., 2018. Gene co-expression analysis for functional classification and gene–disease predictions. *Briefings in bioinformatics*, 19(4), pp.575-592.
- Wainberg, M., Alipanahi, B. and Frey, B.J., 2016. Are random forests truly the best classifiers?. *Journal of Machine Learning Research*, 17(110), pp.1-5.
- Yao, L., Chi, Y., Hu, X., Li, S., Qiao, F., Wu, J. and Shao, Z.M., 2016. Elevated expression of RNA methyltransferase BCDIN3D predicts poor prognosis in breast cancer. *Oncotarget*, 7(33), p.53895.
- Zeng, Q., Cao, J., Niu, Y., Zhao, X., Wang, Y., Liu, W., Liu, B., Chen, Y., Fan, Y. and Dou, J., 2023. Identification of recurrence-related mRNAs and noncoding RNAs in hepatocellular carcinoma following liver transplantation. *The Turkish Journal of Gastroenterology*, 34(4), p.394.
- Zhang, W., Yu, Y., Hertwig, F., Thierry-Mieg, J., Zhang, W., Thierry-Mieg, D., Wang, J., Furlanello, C., Devanarayan, V., Cheng, J. and Deng, Y., 2015. Comparison of RNA-seq and microarray-based models for clinical endpoint prediction. *Genome biology*, 16, pp.1-12.

Appendix

Table S1. Clinical data used in this study.

	ID	Gender	Age	clinico.genetic.subgroup	MYCN.status	INSS.Stage	HighRisk	Progression	DeathFromDisease
0	ICGC_001	M	987.0	ST4	1	4	HR	1	1
1	ICGC_002	M	1808.0	ST4	1	4	HR	1	1
2	ICGC_003	F	625.0	MNA	Amp	4	HR	1	1
3	ICGC_004	F	335.0	MNA	Amp	2b	HR	0	0
4	ICGC_005	F	536.0	MNA	Amp	4	HR	1	1
...	...	...	...	...	...	...	...	...	...
368	ICGC_494	M	56.0	ST1	1	1	LR	0	0
369	ICGC_495	M	163.0	ST1	1	1	LR	0	0
370	ICGC_496	M	132.0	ST1	1	1	LR	1	0
371	ICGC_497	F	379.0	ST1	1	1	LR	0	0
372	ICGC_498	F	530.0	ST1	1	1	LR	0	0

373 rows × 9 columns

Table S2. Definition of 4 major clinico-genetic neuroblastoma subgroups (reproduced from Zhang et al, 2015).

Subgroup	MYCN status	INSS stage	Age at diagnosis	N
MNA	Amplified	Any	≤18 and >18 months	92
ST1	Normal	Stage 1	≤18 and >18 months	117
ST4	Normal	Stage 4	≤18 and >18 months	116
ST4S	Normal	Stage 4S	≤18 months	48

Table S3. Loadings from PCA analysis of input features, including gene and clinical data.

	PC1	PC2
Age	9.998055e-01	0.012620
14416.0	3.697973e-04	0.000439
39308.0	3.634551e-04	0.002011
10433.0	-3.611733e-04	0.001832
5490.0	-3.600668e-04	0.003656
...	...	...
32333.0	-3.144140e-08	0.006074
16277.0	2.769628e-08	0.003542
26412.0	2.599310e-08	0.004942
22666.0	-5.482048e-09	0.005951
13673.0	1.905577e-09	0.004960

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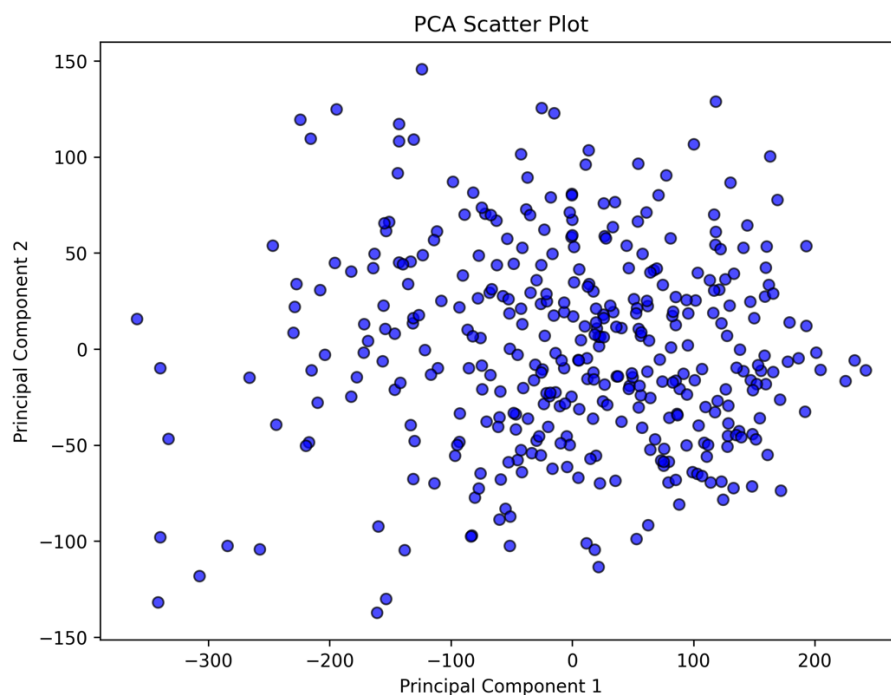


Figure S1. Principle Component Analysis of full microarray dataset.

Data S1

Output from training for other clinical endpoints.

High Risk

Final Test Accuracy - SVM: 0.8800

Final Test Accuracy - RF: 0.9200

Final Selected Features: {'34105.0', '1995.0', '32099.0', '2731.0', '34728.0', '14867.0', '23272.0', '3150.0', '9458.0', '23224.0', '45207.0'}

Best SVM parameters on selected features: {'C': 10, 'kernel': 'rbf'}

Best RF parameters on selected features: {'max_depth': 10, 'min_samples_split': 10, 'n_estimators': 200}

Progression

Final Test Accuracy - SVM: 0.7333

Final Test Accuracy - RF: 0.7067

Final Selected Features: {'17520.0', '4881.0', '28655.0'}

Best SVM parameters on selected features: {'C': 0.1, 'kernel': 'sigmoid'}

Best RF parameters on selected features: {'max_depth': 20, 'min_samples_split': 10, 'n_estimators': 200}

INSS Stage

Final Test Accuracy - SVM: 0.9333

Final Test Accuracy - RF: 0.9333

Final Selected Features: {'clinico.genetic.subgroup_ST1', '35733.0', '10772.0', 'clinico.genetic.subgroup_ST4', '22495.0', '2731.0', '34728.0', '14867.0', '39434.0', '8310.0', '2396.0', 'clinico.genetic.subgroup_ST4S', '9458.0', '23224.0', '44212.0'}

Best SVM parameters on selected features: {'C': 1, 'kernel': 'rbf'}

Best RF parameters on selected features: {'max_depth': None, 'min_samples_split': 2, 'n_estimators': 50}